

Project Design Phase Solution Architecture

Date	03 October 2026
Team ID	SWTID-2026-5622
Project Name	Script-Controlled ACL – Restrict Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

The **Script-Controlled ACL** solution is designed to provide dynamic and field-level access control based on the value of a particular field in a record. Instead of providing access only according to a user's role or static permissions, the system uses a script-based Access Control List (ACL) to evaluate field values and determine whether the user should be allowed to access the record or field.

Its goals are to:

- **Restrict unauthorized access** by checking field values before allowing users to access protected information.
- **Implement dynamic access control** using scripted ACL conditions rather than relying only on static roles.
- **Protect sensitive field information** based on predefined business rules.
- **Improve security and data confidentiality** by allowing access only when the required conditions are satisfied.
- **Provide flexible access control** that can be modified according to changing business requirements.
- **Define the security logic, implementation process, and access requirements** for the proposed solution.

Solution Architecture Flow

User → Application → ACL Rule → Script Validation → Field Value Check → Access Decision

Working:

1. The **User** attempts to access a record or protected field.
2. The **Application** receives the access request.
3. The **ACL Rule** is triggered for the requested record/field.
4. The **Script** checks the required field value and other defined conditions.
5. If the field value satisfies the configured condition, **access is granted**.
6. If the condition is not satisfied, **access is denied**.
7. The system therefore ensures that sensitive information is available only to authorized users under the specified conditions.

Architecture Components:

Component	Description
User	Requests access to a record or protected field.
Application/UI	Provides the interface through which the user accesses the data.
ACL	Controls whether the requested data can be accessed.
Script	Contains the dynamic access-control logic.
Field Value	The value used by the script to determine access.
Access Decision	Grants or denies access based on the evaluated condition.
Database/Record	Stores the information that is being protected.

Architecture flow:

User → Application → ACL → Script → Field Value Validation → Grant/Deny Access → Protected Record

Solution Architecture Diagram:



